

MIAE 221: Materials Science for Engineers

Lecture 6: Crystal Structures 3 — Atomic Densities, XRD & Anisotropy (Explained)

Concordia University · Department of Mechanical, Industrial & Aerospace Engineering

Teacher Material: Lecture 6 (Prof. Mamoun Medraj) · Correlated with Callister Chapter 3

Executive Overview & Core Concepts

In solid-state materials engineering, macroscopic mechanical, physical, and optical behavior stems directly from how atoms populate specific spatial lines and crystallographic planes. This explained lecture guide covers the rigorous foundations of:

1. **Linear and Planar Atomic Densities:** Quantitative derivations of atom packing along direction vectors and planes.
 2. **Directional Anisotropy vs. Polycrystalline Isotropy:** Why elasticity varies with direction in single crystals (e.g., turbine blades) while bulk polycrystalline metals behave isotropically.
 3. **Interplanar Spacing & X-Ray Diffraction (XRD):** The wave-optics principles of constructive interference and the mathematical derivation of Bragg's Law.
 4. **Powder Diffractometry & Crystal Identification:** How diffraction spectra and selection rules reveal unknown lattice parameters and crystal structures.
 5. **Polymorphism & Allotropy:** Temperature/pressure-induced structural phase shifts in carbon and iron.
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1. Linear Atomic Density (LD)

Definition & Physical Principle

Linear Density (LD) is the number of atom diameters intercepted per unit length along a specific crystallographic direction vector:

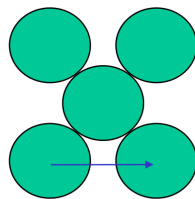
$$LD = \frac{\text{Number of atomic diameters centered on the direction vector}}{\text{Length of the direction vector}} = \frac{n}{L}$$

- **Counting Rule:** For an atom to be counted toward linear density, its **center must lie directly on the direction vector**.
 - End atoms whose centers terminate a line segment inside a unit cell contribute $\frac{1}{2}$ atom each.
 - Atoms whose centers lie strictly along the interior of the line segment contribute 1 full atom each.
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Linear Density

- Calculate the linear density of the [100] direction for the FCC crystal



$$L_D = n/L_L \quad \text{linear density}$$

$$n = 1 \quad \text{atoms}$$

$$L_L = a \quad \text{line length}$$

For FCC $a = \dots\dots\dots$

$$L_D = \dots\dots\dots$$

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Figure 6.1: Linear density calculation along the [100] direction in a Face-Centered Cubic (FCC) unit cell (Dr. Medraj MIAE 221 Lecture 6, Slide 4 & Callister Fig. 3.11). Atoms are centered only at the two cube corners along the edge length a .

- Direction Vector Length:** The direction [100] lies along the edge of the cubic unit cell.

$$L = a$$

In FCC, the lattice parameter relates to atomic radius R by $a\sqrt{2} = 4R \implies a = 2\sqrt{2}R$.

$$L = 2\sqrt{2}R$$

- Atoms Centered on the Vector:** The vector passes through two corner atoms. The face-center atoms do not lie on the edge.

$$n = 2 \times \frac{1}{2} = 1 \text{ atom}$$

- Linear Density Formula:**

$$LD_{[100]} = \frac{1}{a} = \frac{1}{2\sqrt{2}R}$$

- Numerical Example (FCC Aluminum):** Given atomic radius $R = 0.143 \text{ nm} = 0.143 \times 10^{-7} \text{ cm}$:

$$a = 2\sqrt{2}(0.143 \text{ nm}) = 0.4045 \text{ nm}$$

$$LD_{[100]} = \frac{1}{0.4045 \text{ nm}} = 2.47 \text{ nm}^{-1} = 2.47 \times 10^7 \text{ cm}^{-1}$$

Close-Packed Direction: FCC [110]

Along the $[110]$ face diagonal:

- Atoms touch continuously: $L = 4R$.
- Atom count: $2 \text{ corners} \times \frac{1}{2} + 1 \text{ face center} = 2 \text{ atoms}$.
- Linear Density:

$$LD_{[110]} = \frac{2}{4R} = \frac{1}{2R}$$

- For Aluminum: $LD_{[110]} = \frac{1}{2(0.143 \text{ nm})} = \mathbf{3.50 \text{ nm}^{-1}}$.
 - **Physical Insight:** $LD_{[110]} > LD_{[100]}$ ($3.50 \text{ nm}^{-1} > 2.47 \text{ nm}^{-1}$). The $[110]$ direction is the **close-packed direction** in FCC; this is why plastic deformation (dislocation glide) occurs preferentially along $\langle 110 \rangle$.
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2. Planar Atomic Density (PD)

Definition & Physical Principle

Planar Density (PD) is the number of atoms centered on a crystallographic plane per unit area of that plane:

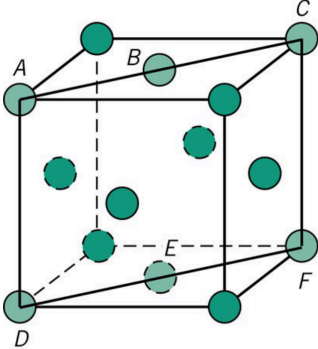
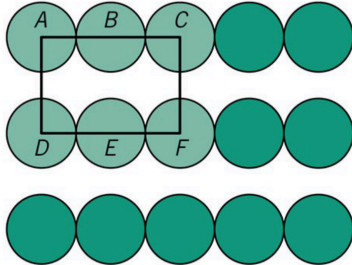
$$PD = \frac{\text{Number of atoms centered on the plane}}{\text{Area of the plane}} = \frac{n}{A}$$

- **Centering Law:** Only atoms whose centers lie strictly within the geometric boundary of the plane are counted.
 - Corner atoms of a planar polygon contribute fractional areas based on their interior plane angles.
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Planar Density

- Calculate the planar density of the (110) plane for the FCC crystal

- Compute planar area
- Compute number of atoms

For an atom to be counted, it has to be centered on that plane.

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Figure 6.2: Planar density geometry for the (110) plane in an FCC crystal (Dr. Medraj Lecture 6, Slides 5–6 & Callister Fig. 3.12). The plane intersects the unit cell as a rectangle of dimensions a by $a\sqrt{2}$.

1. Planar Geometry & Area: The (110) plane cuts diagonally across the FCC unit cell, forming a rectangle:

- Vertical height = a
- Horizontal width = Face diagonal = $a\sqrt{2}$

$$\text{Area } A = a \times a\sqrt{2} = a^2\sqrt{2}$$

Substituting $a = 2\sqrt{2}R$:

$$A = (2\sqrt{2}R)^2\sqrt{2} = 8R^2\sqrt{2} = 8\sqrt{2}R^2$$

2. Number of Centered Atoms (n):

- 4 corner atoms: each interior corner angle of the rectangle is 90° ($\frac{90^\circ}{360^\circ} = \frac{1}{4}$ of a circle).

$$4 \times \frac{1}{4} = 1 \text{ atom}$$

- 2 face-centered atoms: located on the top and bottom faces, each halved by the boundary edge of the rectangle.

$$2 \times \frac{1}{2} = 1 \text{ atom}$$

- Total centered atoms:

$$n = 1 + 1 = 2 \text{ atoms}$$

3. Planar Density Formula:

$$PD_{(110)} = \frac{n}{A} = \frac{2}{8\sqrt{2}R^2} = \frac{1}{4\sqrt{2}R^2}$$

4. Numerical Example (FCC Aluminum): With $R = 0.143 \text{ nm}$:

$$PD_{(110)} = \frac{1}{4\sqrt{2}(0.143 \text{ nm})^2} = \frac{1}{4(1.4142)(0.02045 \text{ nm}^2)} = \mathbf{8.64 \text{ atoms/nm}^2}$$

Engineering Significance of LD and PD

1. Slip Systems & Plastic Deformation:

- Dislocation motion requires overcoming lattice friction (Peierls-Nabarro stress $\tau_{PN} \propto \exp(-2\pi d/b)$).
- Planes with the **highest planar density** have the widest interplanar spacing (d), minimizing lattice resistance.
- Consequently, plastic slip occurs along **directions of highest linear density** within **planes of highest planar density**:
 - **FCC Slip System:** $111\langle 110 \rangle$ (12 slip systems $\implies$ high ductility).
 - **BCC Slip System:** $110\langle 111 \rangle$ (48 active slip systems at elevated temperatures).

2. Speed of Sound: Sound waves propagate via phonon atomic collisions; acoustic velocities are highest along densely packed directions.

3. Surface Energy & Catalysis: Low planar density planes expose more unsatisfied ("dangling") chemical bonds, resulting in higher surface energy and heightened catalytic reactivity.

3. Directional Anisotropy vs. Polycrystalline Isotropy

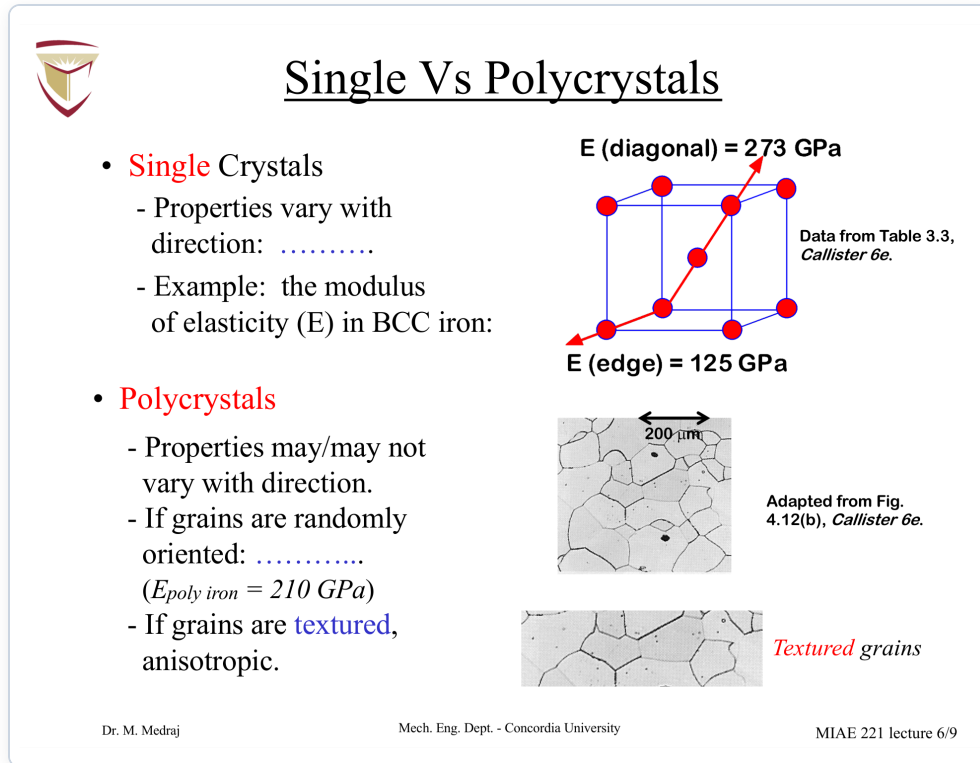


Figure 6.3: Directional dependence (anisotropy) of the Elastic Modulus E in single-crystal BCC Iron (Dr. Medraj Lecture 6, Slides 8–9 & Callister Fig. 3.21). Left: 3D spatial stiffness surface showing dramatic variation between $[111]$ and $[100]$. Right: Polycrystalline aggregate exhibiting microscopic grain boundaries and macroscopic isotropy.

Single Crystals & Anisotropy

In a **single crystal**, the periodic arrangement of atoms is continuous across the entire volume without interruption:

- Physical properties (stiffness, electrical conductivity, magnetic susceptibility, refractive index) depend on the crystallographic direction along which they are measured. This directionality is called **anisotropy**.
- **BCC Iron (Fe) Case Study:**
 - $[111]$ direction (body diagonal, close-packed): $E_{[111]} = 272.7 \text{ GPa}$ (stiffest direction).
 - $[110]$ direction (face diagonal): $E_{[110]} = 210.5 \text{ GPa}$.
 - $[100]$ direction (cube edge, most open): $E_{[100]} = 125.0 \text{ GPa}$ (most compliant direction).
 - Ratio: $\frac{E_{[111]}}{E_{[100]}} = 2.18!$ A single crystal of iron is more than twice as stiff along its body diagonal as along its cube edge.

High-Performance Single Crystal Applications

- **Turbine Blades:** Modern aerospace jet engines operate at temperatures exceeding 1100°C . High-temperature creep occurs primarily via atomic diffusion along grain boundaries. By casting turbine blades as **single crystals** (e.g., PWA 1480, CMSX-4 nickel-base superalloys) oriented along the stiff $[001]$ growth direction, grain boundaries are eliminated, preventing creep failure.
- **Diamond Abrasives:** Industrial cutting tools exploit diamond's maximum hardness along specific crystallographic planes.

Polycrystalline Aggregates & Isotropy

- Most engineering metals are **polycrystals** consisting of millions of microscopic grains separated by grain boundaries.
- If the grains are randomly oriented (equiaxed structure), directional variations average out over macroscopic dimensions, yielding **macroscopic isotropy**.
- When metals undergo heavy plastic deformation (e.g., cold rolling of sheet steel), grains develop a preferred crystallographic orientation (**texture**), reintroducing macroscopic anisotropy.

4. Interplanar Spacing (d_{hkl})

In any crystal lattice, parallel atomic planes designated by Miller indices (hkl) are separated by a constant perpendicular distance called the **interplanar spacing** (d_{hkl}).

For **cubic crystal systems** ($a = b = c$):

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

Key Geometric Relationships

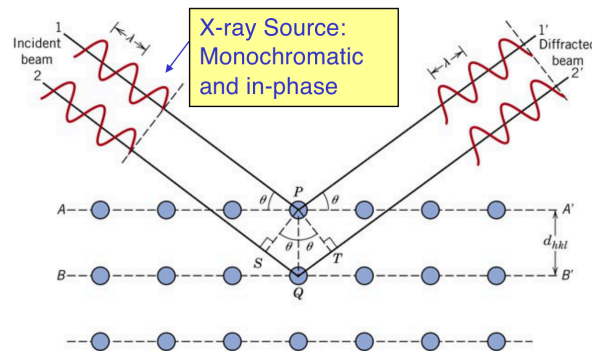
- Higher Miller index planes have **smaller** interplanar spacings.
- For a cubic unit cell with lattice parameter a :
 - $d_{100} = \frac{a}{\sqrt{1^2+0^2+0^2}} = a$
 - $d_{110} = \frac{a}{\sqrt{1^2+1^2+0^2}} = \frac{a}{\sqrt{2}} \approx 0.707a$
 - $d_{111} = \frac{a}{\sqrt{1^2+1^2+1^2}} = \frac{a}{\sqrt{3}} \approx 0.577a$
 - $d_{200} = \frac{a}{\sqrt{2^2+0^2+0^2}} = \frac{a}{2} = 0.500a$

5. X-Ray Diffraction (XRD) & Bragg's Law

X-ray diffraction is the definitive experimental technique used by materials scientists and engineers to identify unknown crystal structures, quantify lattice parameters, and measure residual stresses.



Bragg's Law



The law: For constructive interference, the additional path length $SQ + QT$ must be an integer number of wavelength (λ).

$$n\lambda = SQ + QT = d_{hkl}\sin\theta + d_{hkl}\sin\theta = 2 d_{hkl}\sin\theta$$

$n = 1, 2, 3, \dots$ order of reflection

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Figure 6.4: Derivation of Bragg's Law for X-ray diffraction from parallel crystallographic planes (Dr. Medraj Lecture 6, Slides 11–13 & Callister Fig. 3.25). Constructive interference requires path difference $SQ + QT = 2d_{hkl}\sin\theta = n\lambda$.

Physical Mechanism of Diffraction

- 1. Electromagnetic Wave Character:** X-rays are high-energy electromagnetic radiation with wavelengths ($\lambda \approx 0.05 - 0.2 \text{ nm}$ or $0.5 - 2 \text{ \AA}$) on the same spatial scale as interatomic crystal spacings ($d \approx 0.1 - 0.3 \text{ nm}$).
- 2. Elastic Scattering:** When an incident X-ray beam strikes an atom, its electrons scatter the wave elastically without changing its wavelength.
- 3. Constructive vs. Destructive Interference:**
 - Scattered rays from adjacent atomic planes travel different path lengths.
 - If the scattered waves emerge **out of phase**, destructive interference cancels their amplitude, yielding zero detected intensity.
 - If the scattered waves emerge **in phase**, constructive interference produces a sharp, intense diffracted beam.

Step-by-Step Derivation of Bragg's Law

Refer to Figure 6.4:

- Consider two parallel monochromatic X-ray beams incident at angle θ (the **Bragg angle**) on adjacent crystal planes 1 and 2, separated by interplanar spacing d_{hkl} .
- Ray 2 travels further than Ray 1 by the extra path length:

$$\Delta = \overline{SQ} + \overline{QT}$$

3. From right-triangle geometry:

$$\sin \theta = \frac{\overline{SQ}}{d_{hkl}} \implies \overline{SQ} = d_{hkl} \sin \theta$$

$$\sin \theta = \frac{\overline{QT}}{d_{hkl}} \implies \overline{QT} = d_{hkl} \sin \theta$$

4. Total extra path length:

$$\Delta = 2d_{hkl} \sin \theta$$

5. For constructive interference, this path difference must equal an integer number (n) of complete wavelengths (λ):

$$n\lambda = 2d_{hkl} \sin \theta$$

Where:

- n = Order of reflection ($n = 1, 2, 3 \dots$; typically $n = 1$ in fundamental XRD analysis).
- λ = X-ray wavelength (commonly Copper K_α radiation: $\lambda = 0.1542 \text{ nm} = 1.542 \text{ \AA}$).
- d_{hkl} = Interplanar spacing of diffracting planes (hkl).
- θ = Bragg angle (angle between incident beam and crystallographic plane).

6. The Powder Diffractometer & Reflection Rules

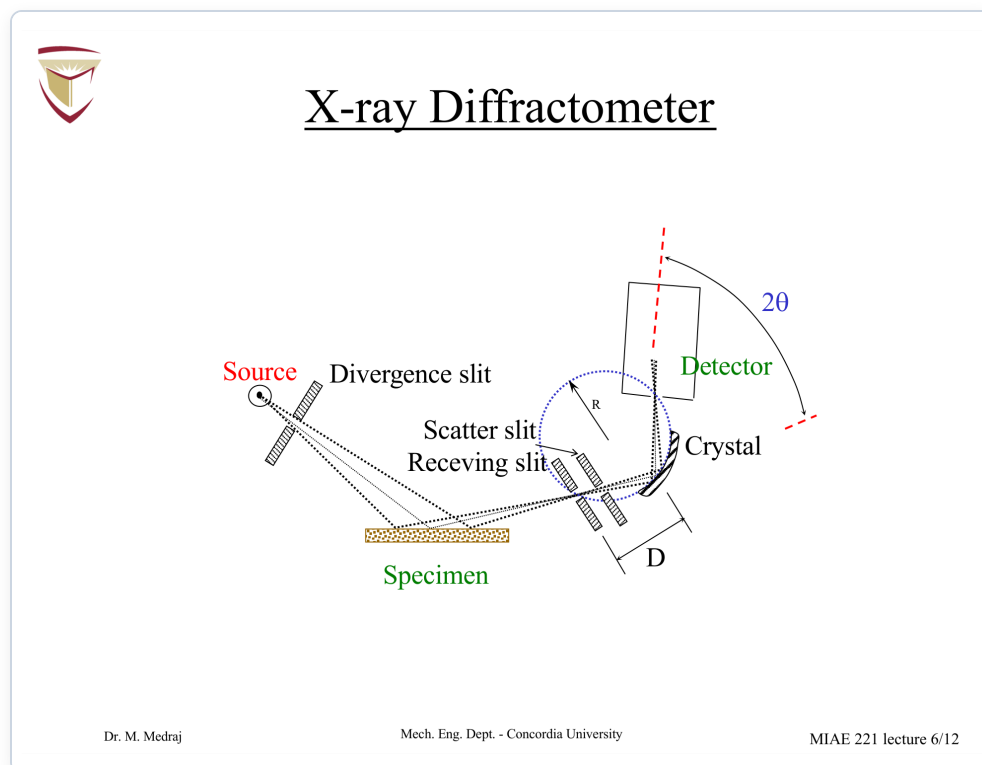


Figure 6.5: Schematic of a Bragg-Brentano $\theta - 2\theta$ powder diffractometer (Dr. Medraj Lecture 6, Slides 14–15 & Callister Fig. 3.26). The X-ray detector rotates at angular velocity 2ω while the specimen stage rotates at ω , recording intensity versus 2θ .

Diffractometer Operation

1. The specimen is ground into a fine powder containing thousands of microscopic crystallites in completely random orientations.
2. At any angle θ , some fraction of crystallites are oriented such that their (hkl) planes satisfy Bragg's Law.
3. The detector rotates at twice the speed of the sample stage (2θ) to maintain the specular reflection condition.
4. **Diffraction Spectrum Output:** The recorder plots scattered X-ray intensity as a function of the **diffraction angle 2θ** (NOT θ).

Selection Rules for Crystal Structure Identification

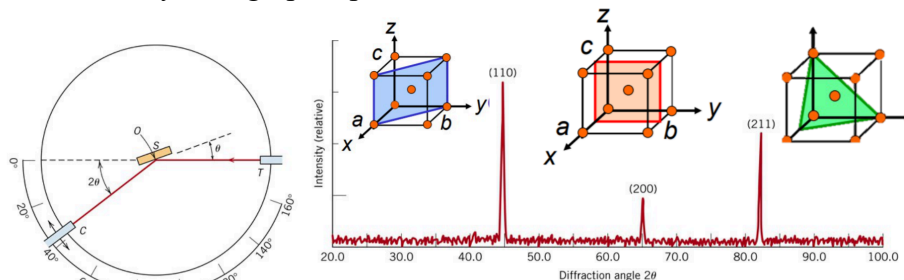
Not all planes produce diffraction peaks; destructive interference between atoms within the unit cell completely extinguishes certain reflections:

Crystal Structure	Diffraction Selection Rule (Condition for Peak to Occur)	Allowed Reflections (hkl)	Forbidden Reflections
BCC	Sum of indices must be even : $h + k + l = 2n$	(110), (200), (211), (220), (310), (222)	(100), (111), (210), (300)
FCC	Indices must be unmixed (all odd OR all even)	(111), (200), (220), (311), (222), (400)	(100), (110), (210), (211)
Simple Cubic	All reflections allowed	All (hkl)	None



Diffractometer Technique

- Use **powder** (or polycrystalline) sample to guarantee some particles will be oriented properly such that every possible set of crystallographic planes will be available for diffraction.



Each material has a unique set of planar distances and extinctions, making X-ray diffraction useful in analysis of an unknown material.

Figure 6.6: Experimental X-ray diffraction spectrum for polycrystalline Face-Centered Cubic (FCC) Aluminum (Dr. Medraj Lecture 6, Slide 17 & Callister Fig. 3.27). Diffraction peaks correspond only to unmixed Miller index planes (111), (200), (220), (311), (222)

Comprehensive Solved Problem: Lattice Parameter from XRD (Dr. Medraj Lecture 6 Example)

Problem Statement: High-purity BCC iron (Fe) is analyzed in a powder diffractometer using monochromatic X-radiation with $\lambda = 0.1790 \text{ nm}$. The first-order reflection ($n = 1$) from the (220) plane family produces a diffraction peak. Given that the lattice parameter of BCC iron is $a = 0.2866 \text{ nm}$:

1. Compute the interplanar spacing d_{220} .
2. Determine the Bragg angle θ and the experimental diffraction angle 2θ .

Solution:

1. **Compute Interplanar Spacing d_{220} :**

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

$$d_{220} = \frac{0.2866 \text{ nm}}{\sqrt{2^2 + 2^2 + 0^2}} = \frac{0.2866 \text{ nm}}{\sqrt{4 + 4 + 0}} = \frac{0.2866}{\sqrt{8}} = \frac{0.2866}{2.8284} = \mathbf{0.1013 \text{ nm}}$$

2. **Compute Bragg Angle θ :** From Bragg's Law ($n = 1$):

$$\lambda = 2d_{220} \sin \theta \implies \sin \theta = \frac{\lambda}{2d_{220}}$$

$$\sin \theta = \frac{0.1790 \text{ nm}}{2(0.1013 \text{ nm})} = \frac{0.1790}{0.2026} = 0.8835$$

$$\theta = \arcsin(0.8835) = \mathbf{62.07^\circ}$$

3. **Compute Experimental Diffraction Angle 2θ :**

$$2\theta = 2 \times 62.07^\circ = \mathbf{124.14^\circ}$$

Exam Trap Alert

: Diffractometer readouts always present the x -axis as 2θ . If an exam question asks for θ , divide the peak position by 2; if it asks for the diffractometer peak position, multiply θ by 2.

7. Polymorphism and Allotropy

Polymorphism is the phenomenon where a material can exist in two or more distinct crystal structures depending on temperature and pressure. When observed in elemental solids, it is termed **allotropy**.



Polymorphism and Allotropy

- Some materials may have more than one **crystal structure** depending on **temperature** and **pressure** - called *POLYMORPHISM*
- Carbon
 - graphite
 - diamond
- Iron BCC and FCC

Figure 6.7: Allotropic phase transformations in engineering solids (Dr. Medraj Lecture 6, Slides 18–19 & Callister Fig. 3.22). Top: Carbon allotropes (diamond tetrahedral sp^3 network vs graphite layered hexagonal sp^2 sheets). Bottom: Temperature-dependent phase transformations in pure iron (α -BCC $\rightarrow$ γ -FCC $\rightarrow$ δ -BCC).

Iron Allotropic Phase Transformations (Atmospheric Pressure)

- **α -Ferrite** (Room Temp up to 912°C): Body-Centered Cubic (BCC), magnetic, low carbon solubility (≤ 0.022 wt).
- **γ -Austenite** (912°C to 1394°C): Face-Centered Cubic (FCC), non-magnetic, dense packing, high carbon solubility (up to 2.14 wt).
- **δ -Ferrite** (1394°C to melting point 1538°C): Body-Centered Cubic (BCC), stable at high temperature.
- **Volume Change during Heat Treatment:** The transformation from BCC α -ferrite to FCC γ -austenite involves a -0.5 **volumetric contraction** because FCC has a higher atomic packing factor ($\text{APF} = 0.74$) than BCC ($\text{APF} = 0.68$). This volumetric contraction and expansion during rapid quenching is the root cause of thermal stresses and quench cracking in steel metallurgy.

Carbon Allotropes

1. Diamond:

- Each carbon atom is covalently bonded to 4 neighbors in a 3D tetrahedral network (sp^3 hybridization).
- Extreme hardness (10 on Mohs scale), high thermal conductivity ($2000 \text{ W/m} \cdot \text{K}$), electrical insulator ($E_g = 5.5 \text{ eV}$).

2. Graphite:

- Carbon atoms form hexagonal planar rings with strong in-plane covalent bonds (sp^2 hybridization).
- Adjacent sheets are bonded only by weak van der Waals forces, allowing easy inter-sheet shearing (solid lubricant, pencil lead). Delocalized π -electrons make graphite an electrical conductor along basal planes.

8. Master Formula & High-Yield Summary Matrix

Property / Concept	Governing Equation	Key Units	Critical Exam Traps
Linear Density (LD)	$LD = \frac{n}{L}$	nm^{-1} or m^{-1}	Only count atoms with centers on the line segment; edge endpoints contribute $\frac{1}{2}$.
Planar Density (PD)	$PD = \frac{n}{A}$	atoms/nm^2 or m^{-2}	Only count atoms with centers inside plane bounds; corners contribute fractional angles.
Interplanar Spacing (Cubic)	$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$	nm or $\text{\AA}$	Forgetting the square root; higher indices yield smaller d .
Bragg's Law	$n\lambda = 2d_{hkl} \sin \theta$	λ, d in same units	Confusing θ with 2θ ; ensuring angle mode in calculator is degrees.
BCC Selection Rule	$h + k + l = \text{even}$	Dimensionless	Reflections like (111) are forbidden in BCC ($1 + 1 + 1 = 3$, odd).
FCC Selection Rule	h, k, l all odd or all even	Dimensionless	Reflections like (110) are forbidden in FCC (mixed parity).
Volume Contraction $\alpha \rightarrow \gamma$	$\Delta V < 0$ (BCC $\rightarrow$ FCC)		FCC (APF = 0.74) is denser than BCC (APF = 0.68).